Problem 4: Find if Doubly Linked List is a Palindrome (easy)

Problem Statement

Given a doubly linked list, determine whether it is a palindrome.

A doubly linked list is a palindrome if it reads the same backward as forward, utilizing the previous and next pointers of the nodes.

Examples

1. **Example 1**:
   - **Input**: 1 <-> 2 <-> 3 <-> 2 <-> 1
   - **Output**: true
   - **Justification**: The list reads the same backward as forward.
2. **Example 2**:
   - **Input**: 1 <-> 2 <-> 2 <-> 3
   - **Output**: false
   - **Justification**: Reading backward, the list is 3 <-> 2 <-> 2 <-> 1, which is not the same as reading forward.
3. **Example 3**:
   - **Input**: 1 <-> 1 <-> 1 <-> 1
   - **Output**: true
   - **Justification**: All elements are the same, so the list is a palindrome.

Try it yourself

Try solving this question here:

Java

```
// class DLNode {
```

```
//     int val;
//     DLNode next, prev;
//     DLNode(int val) { this.val = val; }
// }

class Solution {
    public boolean isPalindrome(DLNode head) {
        // ToDo: Write your code here.
        return false;
    }
}

```

Test Case | Run Code Result | Stdout

```
[1, 2, 1]
```

Use Example Testcases | Run | Submit

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